ARTI 403 – Image Processing | Lab 6

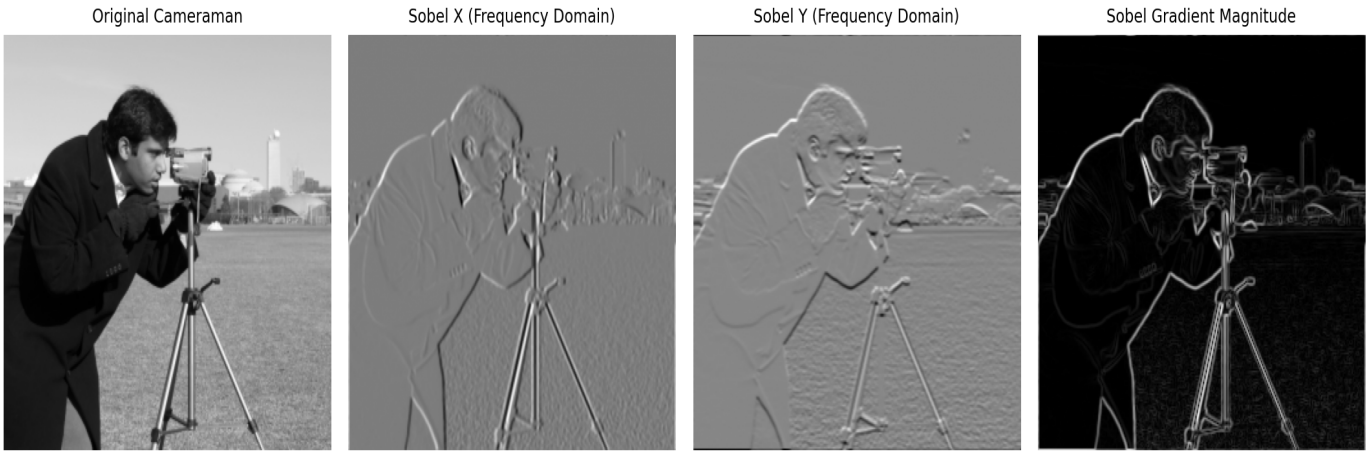
Frequency Domain: Sobel Filtering

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Objective

Apply Sobel X and Sobel Y filters to the Cameraman image in the frequency domain and show the results.

Result image



Method

1. Load and resize the grayscale Cameraman image to 256 × 256.
2. Center-embed each 3 × 3 Sobel kernel in an image-sized array and apply FFT after ifftshift.
3. Compute the image FFT and multiply it by each filter FFT.
4. Apply inverse FFT to obtain Sobel X and Sobel Y responses; combine them using gradient magnitude.

Observation and conclusion

Sobel X emphasizes vertical edges, while Sobel Y emphasizes horizontal edges. Gradient magnitude combines the two directional responses and highlights edges in either direction. The results show frequency-domain edge detection on the Cameraman image.

Code used to generate the result

```
import numpy as np
import matplotlib.pyplot as plt
from skimage import data
from skimage.transform import resize

image = resize(data.camera(), (256, 256), anti_aliasing=True)
image = np.asarray(image, dtype=float)

sobel_x = np.array([[-1, 0, 1], [-2, 0, 2], [-1, 0, 1]], dtype=float)
sobel_y = np.array([[-1, -2, -1], [0, 0, 0], [1, 2, 1]], dtype=float)

def center_embed_kernel(kernel, shape):
    padded = np.zeros(shape, dtype=float)
    kh, kw = kernel.shape
    ph, pw = shape
    cy, cx = ph // 2, pw // 2
    padded[cy-kh//2:cy-kh//2+kh, cx-kw//2:cx-kw//2+kw] = kernel
    return np.fft.fft2(np.fft.ifftshift(padded))

F = np.fft.fft2(image)
G_x = np.fft.ifft2(F * center_embed_kernel(sobel_x, image.shape)).real
G_y = np.fft.ifft2(F * center_embed_kernel(sobel_y, image.shape)).real
gradient_magnitude = np.hypot(G_x, G_y)

plt.figure(figsize=(12, 4))
for i, (result, title) in enumerate([
    (image, "Original Cameraman"),
    (G_x, "Sobel X (Vertical Edges)"),
    (G_y, "Sobel Y (Horizontal Edges)"),
    (gradient_magnitude, "Gradient Magnitude")
], 1):
    plt.subplot(1, 4, i)
    plt.imshow(result, cmap="gray")
    plt.title(title)
    plt.axis("off")
plt.tight_layout()
plt.show()
```